



# Structural Organisation - 100 MCQs

NCERT Nichod : Biology Practice Series NEET 2025



Old NCERT  
+  
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# Structural Organisation in Animal (Tissue + Frog + Cockroach)



**BIOLOGYLIVE**



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

1. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

**Assertion (A) :** Areolar tissue present beneath skin and it is a connective tissue. ✓

**Reason (R) :** In areolar tissue fibre and fibroblast are <sup>loosely.</sup> ~~compactly~~ packed. ✗

In the light of the above statements, choose the correct answer from the options given below :

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

✓ (2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

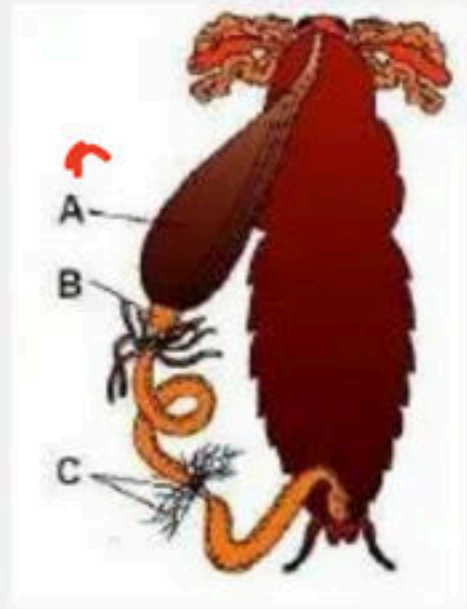
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

## 2. Identify A,B,C in diagram :



A: crop  
B: Gizzard  
C: - M.T

- (1) A - Hepatic Caeca , B - Crop, C - Malpighian tubules
- ✓(2) A - Crop , B - Gizzard , C - Malpighian tubules
- (3) A - Gizzard , B - Crop , C - Hepatic Caeca
- (4) A - Crop , B - gastric ceaca, C - Malpighian tubules



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

3. Match Column- I with Column- II and select the correct option :

Column - I

Column-II

Epithelial tissue

Location

a. Cuboidal

b. Ciliated

c. Columnar

d. Squamous

i. Wall of blood vessels

ii. Lining of stomach and intestine

iii. Inner lining of fallopian tube

iv. Duct of glands

(1) a -i, b-ii, c-iii, d- iv

(2) a -ii, b-iii, c-i, d- iv

✓ (3) a -iv, b-iii, c-ii, d- i

(4) a -iii, b-iv, c-i, d- ii





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

4. Match Column-I with Column -II and choose the correct option:

## Column - I

a. Loose connective

b. Dense regular

c. Dense irregular

d. Specialised connective

## Column-II

i. Tendons and ligaments tissue

ii. Skin connective tissue

iii. Cartilage, bones, blood connective tissue

iv. Fibroblasts, tissue macrophages and mast cells

(1) a -i, b-iv, c-ii, d- iii

(3) a -iv, b-i, c-ii, d- iii

(2) a -i, b-iv, c-iii, d- ii

(4) a -iv, b-ii, c-i, d- iii





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

5. Epithelium, mainly present in the inner surface of hollow organs like bronchioles and fallopian tubes:

(1) Columnar epithelium

(2) Cuboidal epithelium

✓(3) Ciliated epithelium

(4) Squamous epithelium



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

6. In frog the gaseous exchange during hibernation and aestivation takes place through :

- (1) Buccal cavity
- (2) Lungs and Buccal cavity
- ✓(3) Skin = cutaneous
- (4) Lungs

↓  
Skin





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

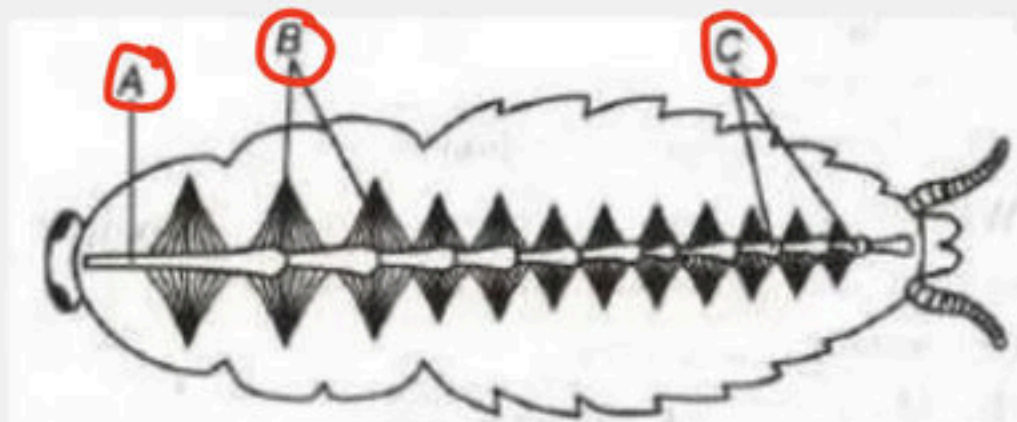
7 In Frog RBC's are : v.v.v. Imp

- (1) Anucleated with Haemoglobin
- (2) Nucleated without Haemoglobin
- (3) Anucleated without Haemoglobin
- ✓ (4) Nucleated with Haemoglobin



## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

8. Given below is the figure of open circulatory system of cockroach. Identify A, B, C and choose the correct option.



- (1) A-Lateral aorta, B-Ciliary muscles, C-Chambers of heart
- (2) A-Internal aorta, B-Alary muscles, C-Chambers of heart
- ☒ (3) A-Anterior aorta, B-Alary muscles, C-Chambers of heart
- (4) A-Posterior aorta, B-Fibrous muscles, C-Chambers of heart





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

V-V-IMP

9. Select the correct sequence of organs in the alimentary canal of cockroach starting from mouth

(1) Pharynx → Oesophagus → Gizzard → Crop → Ileum → Colon → Rectum

(2) Pharynx → Oesophagus → Gizzard → Ileum → Crop → Colon → Rectum

(3) Pharynx → Oesophagus → Ileum → Crop → Gizzard → Colon → Rectum

✓✓ (4) Pharynx → Oesophagus → Crop → Gizzard → Ileum → Colon → Rectum



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

<sup>1/1 mp</sup>  
**10** Pronotum use for which sclerite of periplaneta :

- (1) Dorsal sclerite of head
- (2) Ventral sclerite of abdomen
- ☒ (3) Dorsal sclerite of prothorax
- (4) Dorsal sclerite of abdomen





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

11. Identify the given mouth part of cockroach :



(1) Upper lip

✓ (2) Mandible

(3) Hypopharynx

(4) Lower lip



## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

12. Given below are two statements :

Statement I : Heart of cockroach consists of 13 chamber.

Statement-II : Only adult cockroach have wings.

Choose the correct answer from the option given below:

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

13. Which one is part of male reproductive system of cockroach :

- (1) Spermatheca
- (2) Vestibulum
- (3) Collateral gland
- ✓ (4) Ejaculatory ducts



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

14. Six highly chitinous plate like teeth are found in ..... of cockroach :

- (1) Maxilla
- (2) Mandible
- ✓ (3) Gizzard
- (4) Rectum





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

15. The mosaic vision of cockroach has :

2025

- ✓ (1) More sensitivity but less resolution
- (2) Less sensitivity but more resolution
- (3) Both sensitivity and resolution are high
- (4) Both sensitivity and resolution are low



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

16. Which statement is incorrect :

→ *Kynopharynx*

- ✓ (1) Labrum act as tongue in cockroach
- (2) Epithelial tissue has a free surface
- (3) Epithelial tissue provides a covering or a lining for some part of the body
- (4) The tissues are broadly classified into Epithelial, Connective, Muscular and Neural





## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

17. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Blood vascular system of cockroach is an open type.

Reason (R) : Blood vessel are poorly develop and open into space or Haemocoel and visceral organs located into the haemocoel are bathed in blood or haemolymph.

Choose the correct answer from the options given below

- (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (2) (A) is true but (R) is false
- (3) (A) is false but (R) is true
- (4) Both (A) and (R) are true and (R) is the correct explanation of (A)



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

18. Which one is product of ductless gland :

(1) Mucus

(2) Saliva

(3) Milk

☒ (4) Hormone

↳ endocrine





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

v.i.m.p

19. Match the column I and column II and choose correct answer :

Column - I

Column - II

(A) Hepatic caeca

(i) 9 -10

(B) Malpighian tubules

(ii) 13

(C) Oothecae

(iii) 100 – 150

(D) Heart chamber

(iv) 6 – 8

✓ (1) A-iv, B-iii, C-i, D-ii

(2) A-iii, B-iv, C-i, D-ii

(3) A-iii, B-iv, C-ii, D-i

(4) A-iv, B-iii, C-ii, D-i



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

20. Match the column I and column II and choose correct answer :

Column – I

(A) Adipose tissue

(B) Ligament

(C) Cartilage

(D) Blood

Column - II

(i) Skeletal connective tissue

(ii) Dense connective tissue

(iii) Loose connective tissue

(iv) Fluid connective tissue

✓ (1) A-iii, B-ii, C-i, D-iv

(3) A-ii, B-i, C-iii, D-iv

(2) A-iii, B-i, C-ii, D-iv

(4) A-i, B-ii, C-iv, D-iii





## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

21. Given below are two statements :

Statement I : Cardiac muscle tissue is a contractile tissue present only in heart.

Statement-II : Cardiac muscle is an involuntary muscle.

Choose the correct answer from the option given below:

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

22. In cockroach hepatic caecae secrete :

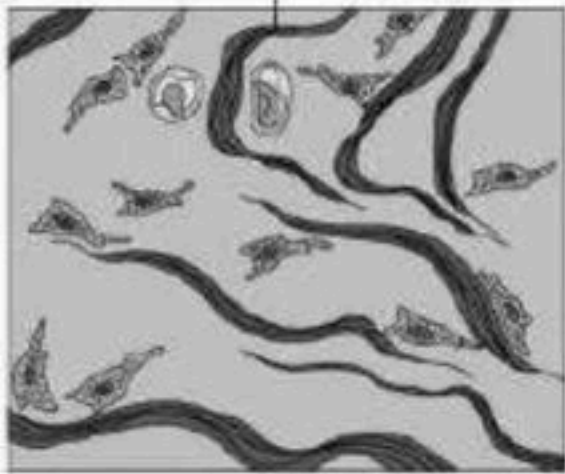
- (1) Scent
- (2) Hormone
- (3) Cellulose
- (4) Digestive juice





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

23. This figure denotes which type of tissue :



=> Skin

- (1) Dense regular connective tissue
- ✓ (2) Dense irregular connective tissue
- (3) Adipose tissue
- (4) Any two of above



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

24. In cockroach chitinous plate of each segment are join to each other by :

- (1) Articular membrane
- (2) Ligament
- (3) Arthrodial membrane
- ✓ (4) Both 1 & 3





## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

25. Given below are two statements :

✓ Statement I : Nuclei in cells of columnar epithelium are located at the base.

✓ Statement-II : Ciliated epithelium is modified columnar or cuboidal cells and bear cilia on their free surface.

Choose the correct answer from the option given below:

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- ✓ (4) Both Statement I and Statement II are correct.



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

26. In cockroach forewings arise from :

(1) Prothorax

✓ (2) Mesothorax

(3) Metathorax

(4) Abdomen





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

27. Match Column- I with Column- II and select the correct option :

Column - I

(Tissues)

Column - II

(Location)

a. Squamous epithelium

b. Cubodial epithelium

c. Columnar epithelium

d. Ciliated epithelium

i. Presents in bronchioles + f.1

ii. Present in air sac of lungs

iii. Present in stomach

iv. Duct of glands

☒ (1) a -ii, b-iv, c-iii, d- i

(3) a -iv, b-iii, c-ii, d- i

(2) a -ii, b-iii, c-i, d- iv

(4) a -iii, b-iv, c-i, d- ii





## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

28. Given below are two statements :

**Statement I :** Frog never drink water but absorb it through the skin.

**Statement II :** In frog skin act as aquatic respiratory organ.

In the light of the above statements, choose the correct answer from the options given below

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- ✓ (4) Both Statement I and Statement II are correct.





## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

29. Which of the following statement is correct for frog:

- ✓ (1) On land buccal cavity, skin and lungs act as respiratory organ.
- (2) On land only buccal cavity act as respiratory organ
- (3) On land only lungs act as respiratory organ
- (4) On land only lungs and buccal cavity act as respiratory organ.



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

30. Respiration by lungs is called :

- ✓ (1) Pulmonary respiration
- (2) Cutaneous respiration
- (3) Branchial respiration
- (4) Buccopharyngeal respiration





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

31. All the exoskelton plates of Periplaneta collectively called as :

(1) Tergite

(2) Sternite

(3) Pleurite

✓ (4) Sclerite



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

32. Which of the following specialised connective tissue:

- (1) Cartilage
- (2) Bone
- (3) Blood
- (4) All of these





## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

33. In cockroach each compound eye consist of how many ommatidia :

(1) 4000

(2) 3000

✓ (3) 2000

(4) 5000



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

34. Match Column- I with Column- II and select the correct option :

Column - I

(Tissues)

a. Areolar tissue

b. Adipose tissue

c. Ligament

d. Bone

Column - II

(Composition)

i. Fat storage area

ii. Osteocytes

iii. Loose connective tissue

iv. Dense regular connective tissue

✓ (1) a -iii, b-i, c-iv, d- ii

(3) a -iv, b-iii, c-ii, d- i

(2) a -ii, b-iii, c-i, d- iv

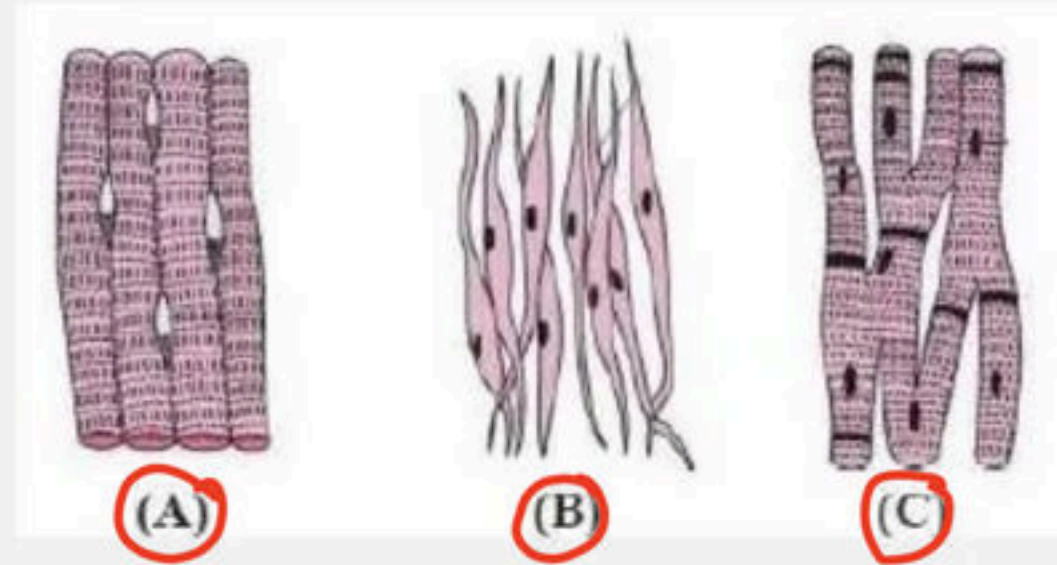
(4) a -iii, b-iv, c-i, d- ii





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

35. Identified the fig. A, B and C and choose the correct option :



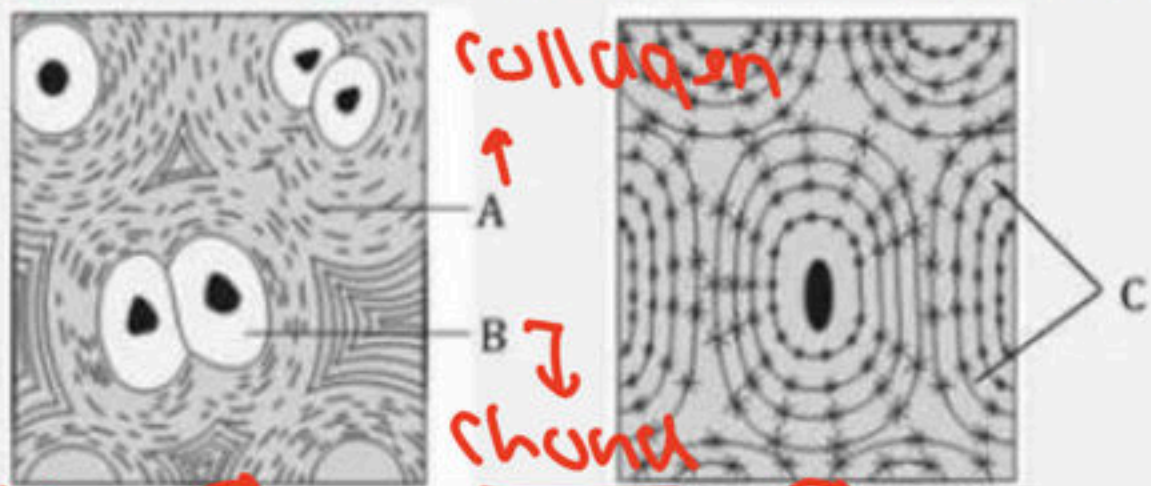
- ☒ (1) A-Skeletal muscle tissue, B-Smooth muscle tissue, C-Cardiac muscle tissue
- (2) A-Skeletal muscle tissue, B-Voluntary muscle tissue, C-Cardiac muscle tissue
- (3) A-Cardiac muscle tissue, B-Skeletal muscle tissue, C-Smooth muscle tissue
- (4) A-Smooth muscle tissue, B-Cardiac muscle tissue, C-Skeletal muscle tissue





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

36. Study the figure I and II carefully and identify the structures A, B and C respectively which are related with specialized connective tissues



cartilage I

II Bone

|     | Fig. I    | Fig. II   | A               | B            | C       |
|-----|-----------|-----------|-----------------|--------------|---------|
| (a) | Bone      | Cartilage | Collagen fibres | Osteoblast   | Lamella |
| (b) | Cartilage | Bone      | Microtubule     | Chondroclast | Lamella |
| (c) | Cartilage | Bone      | Collagen        | Chondroclast | Lamella |
| (d) | Cartilage | Bone      | Collagen        | Chondrocyte  | Lamella |





## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

37. Given below are two statements

Statements I : Ligaments are dense irregular tissue ✗

Statement II : Cartilage is dense regular tissue In the light of the above ✗  
statements, choose the correct answer from the options given below:

- (1) Statement I is false but Statement II is true
- (2) Both Statement I and Statement II are true
- ✓ (3) Both Statement I and Statement II are false
- (4) Statement I is true but Statement II is false



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

38. <sup>Bone</sup> Humerus and muscles are connected with

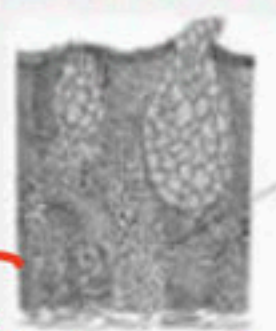
- (1) Ligament (Bone to Bone)
- ✓(2) Tendons
- (3) Both of these
- (4) None of these





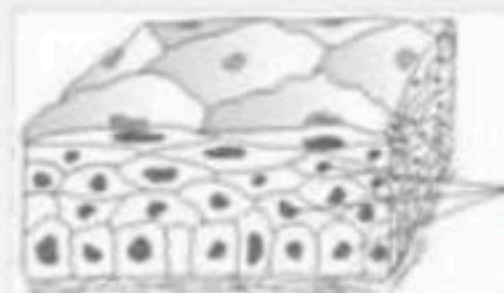
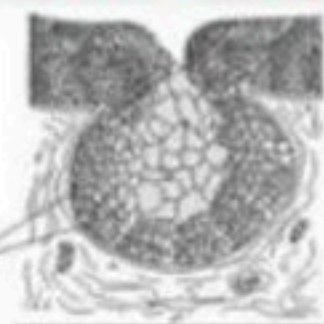
# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

39. See the following figures



A

B



C = compound

Goblet cell

Figure A and B indicate glands while Figure C indicates specific type of tissues. Identify A, B and C

|       | A                   | B                   | C                           |
|-------|---------------------|---------------------|-----------------------------|
| (a)   | Unicellular gland   | Goblet gland        | Pseudostratified epithelium |
| (b)   | Multicellular gland | Unicellular gland   | Pseudostratified epithelium |
| (c)   | Unicellular gland   | Multicellular gland | Pseudostratified epithelium |
| (d) ✓ | Unicellular gland   | Multicellular gland | Compound epithelium         |



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

→ fore wing

40. Tegmina in cockroach, arises from

(1) Metathorax = hind wing

(2) Prothorax and Mesothorax

(3) Prothorax

(4) Mesothorax





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

41. The number of fingers in the hindlimb of frog is

(1) 4 = Forelimb

✓(2) 5

(3) 6

(4) 7



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

42. Consider the following four statements (A – D) related to the common frog Rana tigrina and select the correct option stating which ones are true (T) and which ones are false (F)

Statements :

(A) On dry land it would die due to lack of O<sub>2</sub> if its mouth is forcibly kept closed for a few days (T)

(B) It has four-chambered heart ✗

(C) On dry land it turns uricotelic from ureotelic ✗

(D) Its life-history is carried out in pond water ✓

Options :

|       | (A) | (B) | (C) | (D) |
|-------|-----|-----|-----|-----|
| (a)   | F   | F   | T   | T   |
| (b)   | F   | T   | T   | F   |
| ✓ (c) | T   | F   | F   | T   |
| (d)   | T   | T   | F   | F   |

Amphibia





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

43. Match column-I with column-II and choose the correct option.

| Column I |                              | Column II |                    |
|----------|------------------------------|-----------|--------------------|
| (A)      | <i>Periplaneta americana</i> | (I)       | Hepatic caecae     |
| (B)      | A ring of 6-8 blind tubules  | (II)      | Phylum arthropoda  |
| (C)      | Vascular system              | (III)     | Spiracles          |
| (D)      | 10 pairs of small holes      | (IV)      | Malpighian tubules |
| (E)      | Excretion                    | (V)       | Open type          |

(1) A – I; B – II; C – III; D – IV; E – V

✓ (2) A – II; B – I; C – V; D – III; E – IV

(3) A – II; B – I; C – III; D – V; E – IV

(4) A – III; B – IV; C – II; D – V; E – I



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

44. Match the terms given in column-I with their feature given in column-II and choose the correct option.

| Column I<br>(Terms) |                    | Column II<br>(Features) |                                                                                                      |
|---------------------|--------------------|-------------------------|------------------------------------------------------------------------------------------------------|
| (A)                 | Exocrine gland     | (I)                     | They help to <u>stop</u> substances from leaking across a tissue                                     |
| (B)                 | Endocrine gland    | (II)                    | Hormones are secreted directly into the fluid bathing the gland                                      |
| (C)                 | Tight junctions    | (III)                   | They perform cementing to keep neighbouring cells together                                           |
| (D)                 | Adhering junctions | (IV)                    | Secretes <u>mucus</u> , <u>saliva</u> , earwax, oil, milk, digestive enzymes and other cell products |

✓ (1) A-IV; B-II; C-I; D-III

(2) A-II; B-IV; C-I; D-III

(3) A-IV; B-II; C-III; D-I

(4) A-IV; B-I; C-II; D-III





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

45. Which of the following is not a connective tissue :

v.v. Imp

(1) Cartilage

✓ (2) Neuroglia

(3) Blood

(4) Adipose tissue



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

## 46. Match List - I with List –

|     | List – I       |       | List – II                       |
|-----|----------------|-------|---------------------------------|
| (A) | Bronchioles    | (i)   | Dense Regular Connective Tissue |
| (B) | Goblet cell    | (ii)  | Loose Connective Tissue         |
| (C) | Tendons        | (iii) | Glandular Tissue                |
| (D) | Adipose Tissue | (iv)  | Ciliated Epithelium             |

Choose the correct answer from the options given below

(1)(A) – (ii), (B) – (i), (C) – (iv), (D) – (iii)

(2)(A) – (iii), (B) – (iv), (C) – (ii), (D) – (i)

✓ (3)(A) – (iv), (B) – (iii), (C) – (i), (D) – (ii)

(4)(A) – (i), (B) – (ii), (C) – (iii), (D) – (iv)





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

47. <sup>x</sup> Assertion : Tendon is present in <sup>"</sup>all <sup>"</sup>bone joints. <sup>ligament</sup> <sup>x</sup> Bone - Bone

<sup>x</sup> Reason : Tendon connects the bones at the joints & holds them in position <sup>x</sup>

(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion

(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion

(3) If the assertion is true but the reason is false

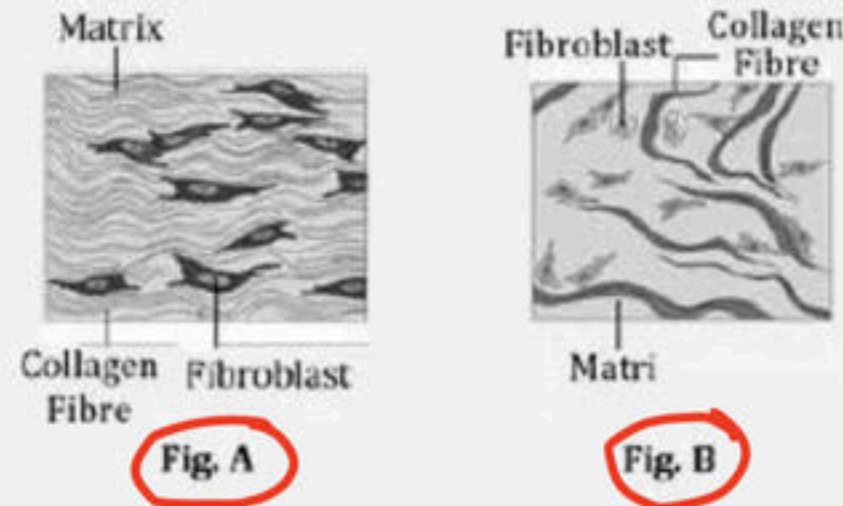
<sup>✓</sup> (4) If both the assertion and reason are false





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

48. Identify the following figure – A and B respectively



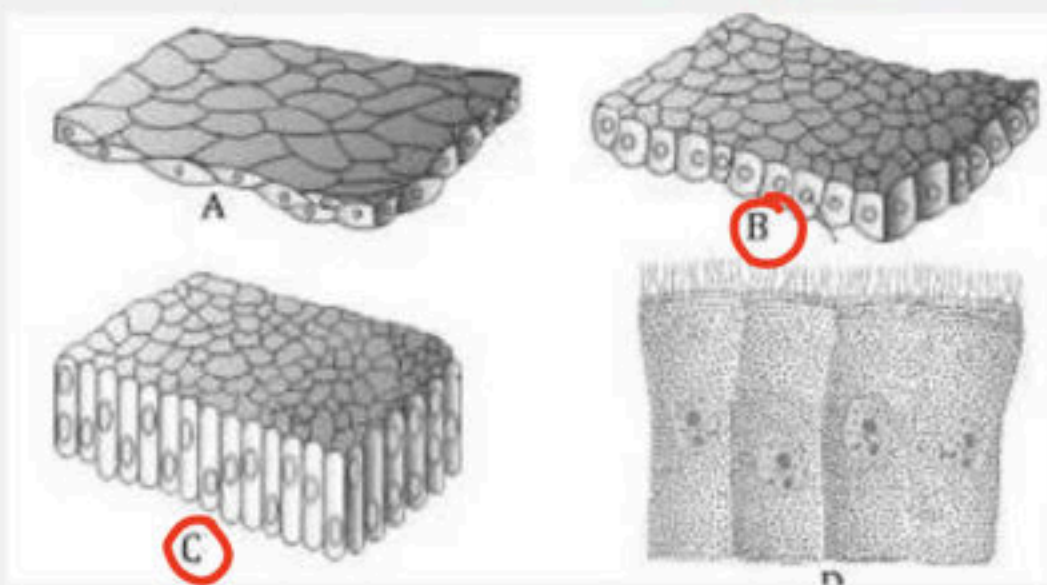
- (1) Connective tissue proper, specialized connective tissue
- (2) Adipose tissue, specialized connective tissue
- (3) Dense irregular connective tissue, dense regular connective tissue
- ✓ (4) Dense regular connective tissue, dense irregular connective tissue





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

49. See the following figure and identify the following simple epithelial tissues :



|      | <b>A</b>                  | <b>B</b> | <b>C</b> | <b>D</b>                             |
|------|---------------------------|----------|----------|--------------------------------------|
| (a)  | Squamous                  | Cuboidal | Columnar | Pseudostratified columnar (ciliated) |
| (b)  | Pseudostratified squamous | Cuboidal | Columnar | Ciliated columnar                    |
| ✓(c) | Squamous                  | Cuboidal | Columnar | Ciliated columnar                    |
| (d)  | Cuboidal                  | Squamous | Columnar | Ciliated columnar                    |



## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

50. How many eggs are laid by a female frog at a time :

(1) 100 – 200

(2) 500 – 1000

✓ (3) 2500 – 3000

(4) 5000 – 6000





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

51. In frog the colour of dorsal side of body is generally:

- ✓ (1) Olive green with dark irregular spots
- (2) Olive green with dark regular spots
- (3) Pale yellow
- (4) Pale yellow with dark irregular spots



## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

52. How many digits are found in fore limb and hind limb of frog respectively:

- ☒ (1) 4 and 5
- (2) 5 and 4
- (3) 2 and 3
- (4) 3 and 2





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

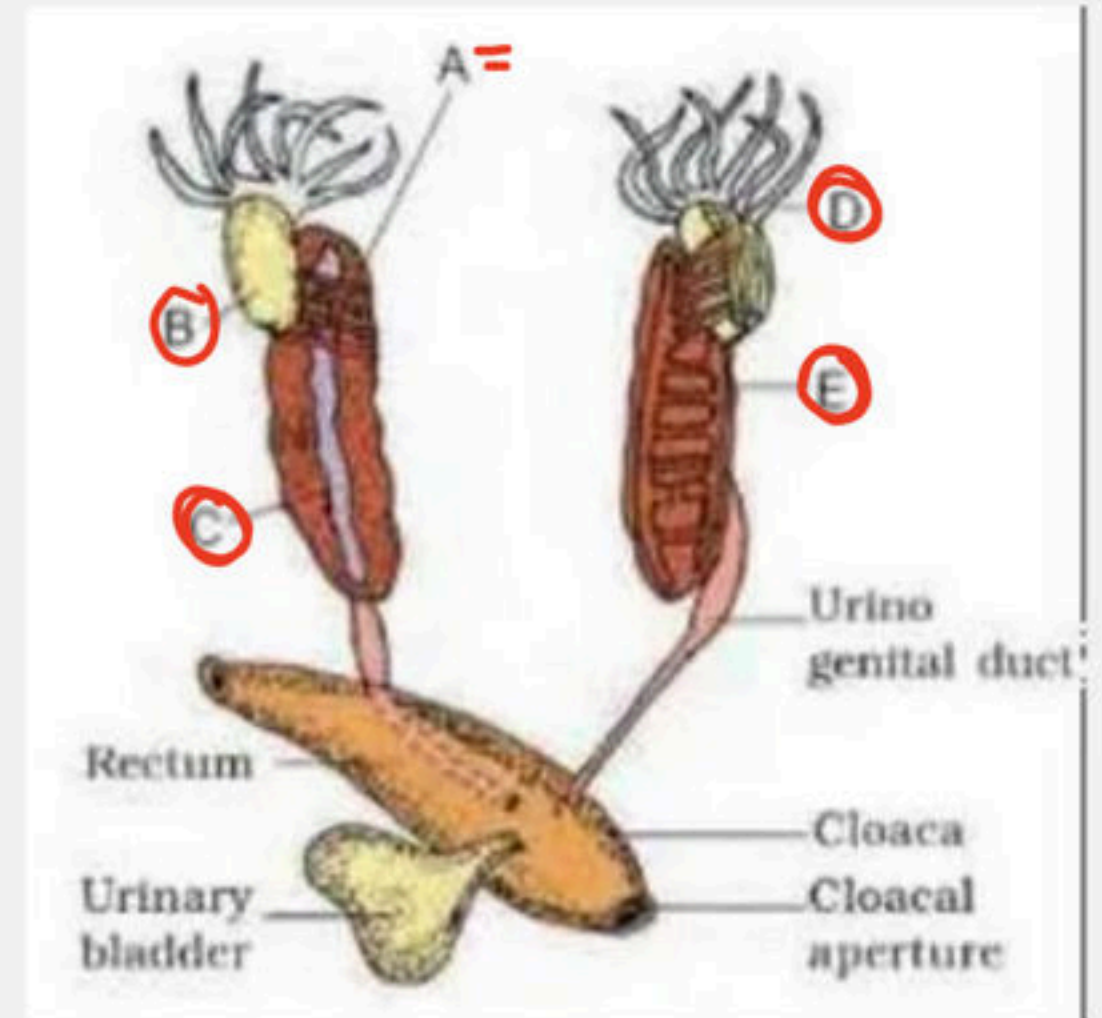
11/11/22  
53. Identify A, B, C and D in the given figure of male reproductive system of frog.

(1) A - Fat bodies, B - Testis, C - Adrenal gland, D - Vasa efferentia, E - Kidney

(2) A - Testis, B - Vasa efferentia, C - Kidney, D - Fat bodies, E - Adrenal gland

✓ (3) A - Vasa efferentia, B - Testis, C - Adrenal gland, D - Fat bodies, E - Kidney

(4) A - Kidney, B - Fat bodies, C - Adrenal gland, D - Testis, E - Vasa efferentia



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

54. Movement of mucus in a specific direction over epithelium is function of :

- (1) Columnar epithelia
- (2) Cuboidal epithelia
- ✓ (3) Ciliated epithelia
- (4) Squamous epithelia





## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

55. Select the correct route for the passage of sperms in male frogs. imp?

- (1) Bidder's canal - Vasa efferentia - Urinogenital duct - Cloaca - Testes - Kidney
- (2) Testes - Cloaca - Kidney - Urinogenital duct - Vasa efferentia - Bidder's canal
- (3) Testes - Vasa efferentia - Bidder's canal - Ureter - Cloaca
- (4) Testes - Vasa efferentia - Kidney - Bidder's canal - Urinogenital duct - Cloaca



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

56. Match the following with column I & column II and choose the correct answer :

Column I (Tissues)

Column II (Present in)

a. Squamous epithelium

i. Cube like cells

b. Cuboidal epithelium

ii. Flattened cells

c. Columnar epithelium

iii. Ciliated cells

d. Ciliated epithelium

iv. Tall and slender like cells

✓ (1) a-ii, b-i, c-iv, d-iii

(2) a-i, b-ii, c-iv, d-iii

(3) a-iii, b-i, c-iv, d-ii

(4) a-i, b-ii, c-iii, d-iv





## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

57. Which is a correct statement :

(a) Ligament is a dense irregular connective tissue ✗

(b) Tendons is a dense irregular connective tissue ✗

(c) Tendons is a dense regular connective tissue ✓

(d) Ligament is a dense regular connective tissue ✓

(1) a & b

(2) b & c

✓ (3) Only c and d

(4) Only d



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

58. The excess of nutrients which are not used immediately are converted into fats and are stored in tissue

(1) Epithelial tissue

✓ (2) Adipose tissue

(3) Neural tissue

(4) All of these





## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

59. A group of similar cells alongwith intercellular substances perform a specific function. Such an organisation is called :

- (1) Organ
- (2) Organ system
- ✓✓ (3) Tissue ✓✓
- (4) Body



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

60. Which one of the following are correct statements for the given diagram :-

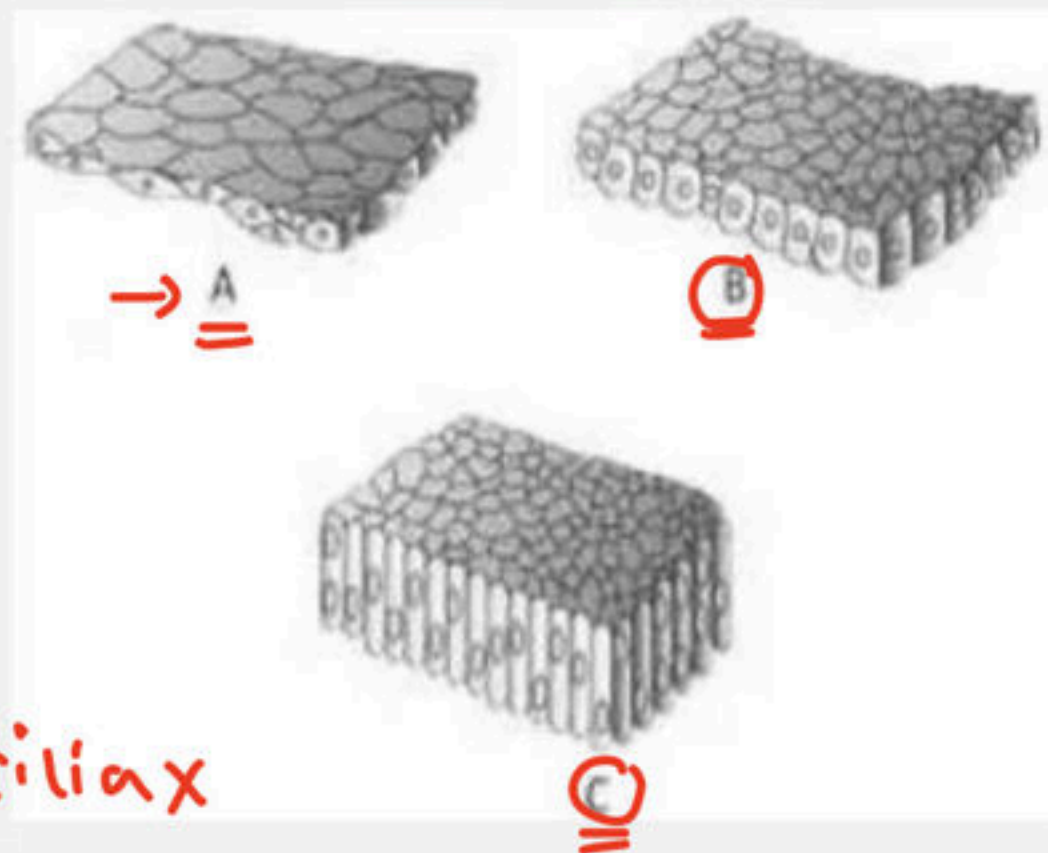
I. A—Irregular boundaries and found in the alveoli of lungs = Squ. ✓

II. B—Found in tubular part of Nephrons ✓

III. C—Help in secretion and absorptions ✓

IV. A—Found in duct of glands and phagocytic ✗

V. C—Their function is to move particles or mucus = cilia ✗



(1) I, II, V

(2) I, II, IV

(3) I, II, III ✓

(4) I, III, V





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

61. The chemical nature of matrix of connective tissue is:

(1) Lipid

✓ (2) Modified polysaccharides → mucopoly

(3) Phospholipids

(4) Steroidal



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

62. Which is a secretion of exocrine gland :

- (1) Milk ✓
- (2) Hormone
- (3) Saliva ✓
- (4) Both 1 and 3 ✓✓

↓  
Enzyme

Endocrine

↓  
Hormone





## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

63. Which is a correct match :

- (1) Glandular epithelium - Tendon
- (2) Exocrine gland - Without duct
- (3) Endocrine gland - With duct
- (4) Connective tissue - Linking and supporting function



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

64. Malpighian tubules in cockroach are present at the junction of

(1) foregut and midgut

✓ (2) midgut and hindgut

(3) foregut and hindgut

(4) midgut and gizzard





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

## 65. Function of neuroglial cell

(1) Protect the ~~nephrons~~ neurons

✓ (2) Support the neuron ✓

(3) Both 1 and 2

(4) Support the nephrons



## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

66. Which of the following statement are correct for smooth muscle fibre

- (1) Smooth muscle fibre taper at both ends
- (2) Smooth muscle fibre taper at only one end
- (3) Smooth muscle fibre do not show striations
- (4) Both 1 and 3





## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

67. In all connective tissue the cells secrete fibres of structural proteins called collagen or elastin except :

(1) Cartilage

(2) Bone

✓ (3) Blood

(4) Adipose



## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

68. Which junctions help to stop substances from leaking across a tissue :

- (1) Gap junctions
- (2) Adhering junctions
- ✓ (3) Tight junctions
- (4) All of the above





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

69. Presence of Nictitating membrane is the characteristic of :

(1) Cockroach

✓ (2) Frog ✓

(3) Earthworm

(4) 1 and 2 both



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

70. With reference to cockroach, match the following columns.

Column - I

Column - II

(Body parts of cockroach) (Location in the body)

a. Anal cerci

i. 4th to 6th segments

b. Tegmina

ii. 10th segment

c. Testes

iii. Forewings

d. Ommatidia

iv. Sclerites

e. Exoskeleton

v. Visual unit

✓ (1) a -ii, b-iii, c-i, d-v, e-iv

(2) a -iv, b-iii, c-ii, d-v, e-i

(3) a -iii, b-iv, c-v, d-ii, e-i

(4) a -v, b-iv, c-iii, d-ii, e-i

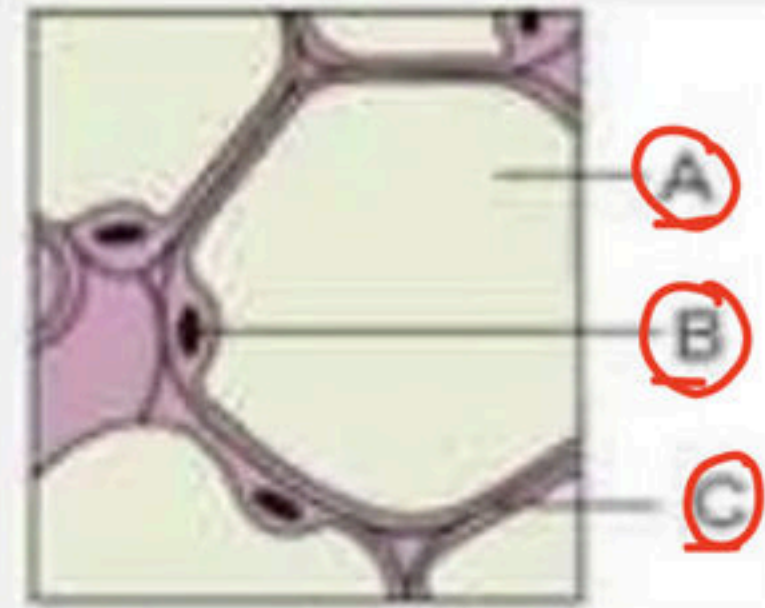




# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

71. Identify A,B,C in diagram :

- ✓ (1) A - Fat storage area, B - Nucleus , C - plasma membrane
- (2) A - Fibroblast, B - Nucleus , C - Macrophage
- (3) A - Fat storage area , B - Fibroblast , C - plasma membrane
- (4) A - Plasma membrane , B - Fibroblast, C - collagen fibres



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

72. Which statement is correct :

- (1) Antennae of cockroach monitor the environment ✓
- (2) In columnar epithelium cells are scale like ✗
- (3) In proximal convoluted tubule, squamous epithelium ✗  
is found
- (4) Compound epithelium is made of single layer of  
cells





## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

73. Given below are two statements :

✓ **Statement I :** Our heart consists of all the four type of tissue epithelial, connective, muscular and neural.

✗ **Statement II :** Inner lining of ducts of Salivary glands and of pancreatic duct made up of compound epithilium.

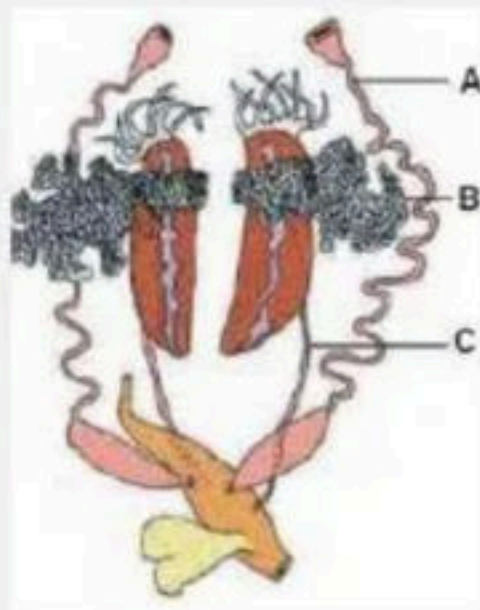
In the light of the above statements, choose the correct answer from the options given below

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- ✓ (4) Both Statement I and Statement II are correct.



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

74. In given figure Identify A, B and C :



A → oviduct

B → ovary

C → ureter

(1) A - Ovary, B - Oviduct, C - Ureter

✓ (2) A - Oviduct, B - Ovary, C - Ureter

(3) A - Ureter, B - Ovary, C - Oviduct

(4) A - Vasa efferentia, B - Testis, C - Ureter





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

75. When two or more organs perform a common function by their physical and chemical interaction, they together form a

(1) Tissue system

✓ (2) Organ system

(3) Tissue

(4) Organ



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

76. Match the following columns with reference to frog.

Column - I

Column - II

(Body parts in frog)

(Location in the body)

a. Respiratory organ

i. Endocrine gland

b. Excretory system

ii. Skin

c. Thymus

iii. Cloaca

d. Brain box

iv. Cranium

e. Nasal epithelium

v. Smell

✓ (1) a -ii, b-iii, c-i, d-iv, e-v

(2) a -i, b-ii, c-iii, d-iv, e-v

(3) a -v, b-iv, c-iii, d-ii, e-i

(4) a -iv, b-iii, c-ii, d-i, e-v





## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

77. Which is a correct statement :

(a) Tight junctions are well developed in cartilage cell X

(b) Neurons are the unit of nervous system ✓

(c) Gap junctions provide no exchange X

(d) Ligament is a dense regular connective tissue ✓✓

(1) Only a & b

(2) Only b & c

(3) Only c

(4) Only b and d ✓



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

78. Which connective tissue is characterized by solid and pliable intercellular material :

- ✓ (1) Cartilage
- (2) Ligament
- (3) Areolar tissue
- (4) Bone





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

79. Given below are two statements :

**Statement I :** Hepatic portal system and Renal portal system both are present in Frog.

**Statement II :** Special venous connection between Kidney and lower part of the body is called Renal portal system.

In the light of the above statements, choose the correct answer from the options given below

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- ✓ (4) Both Statement I and Statement II are correct.



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

80. In cockroach tegmina is :

(1) Metathoracic wing

(2) Hind wing

✓ (3) Forewing

(4) Mouth part





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

81. Head of cockroach is formed by how many segment:

- ☒ (1) 6
- (2) 7
- (3) 8
- (4) 10



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

82. Which segment in female cockroach is boat shaped:

V.V.I.M.P

- (1) 10th sternum
- (2) 6th sternum
- (3) 8th sternum
- (4) 7th sternum





## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

83. In cockroach the testes are situated at which segment of abdomen :

(1) 7th - 9th

(2) 2th - 3th

☒ (3) 4th - 6th

(4) 9th - 10th



84. Correct about intercalated disc :-

- ✓✓ (1) Found in cardiac tissue
- (2) Found at <sup>some</sup> ~~all~~ fusion points ✗
- (3) Both 1 and 2
- (4) None of the above





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

85. Which is not a secretion of exocrine gland :

(1) Ear wax

(2) Mucus

(3) Oil

(4) ☒ Thyroxine = Hormone = Gland = endocrine



## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

86. Refer the given figure and identify the <sup>"</sup>correct characteristic feature.<sup>"</sup>

(i) It is a type of loose connective tissue. ✓

(ii) It contains fibroblast, macrophages, collagen fibres and mast cells. ✓

(iii) The cells of this tissue are specialized to store fats. → Adipose

(iv) The wall of internal organs such as the blood vessels, stomach and intestine contains this type of tissue. ✗

✓ (1) (i) & (ii)

(2) (i) & (iii)

(3) (ii) & (iii)

(4) (iii) & (iv)







# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

87. Which of the following statement(s) is/are correct regarding excretory system of cockroach?

(i) Excretion is performed by malpighian tubules. ✓

(ii) They absorb nitrogenous waste products and convert them into uric acid which is excreted out through hindgut. Hence, this insect called ammonotelic. uricotelic ✓

(iii) In addition, fat body, nephrocytes and uricose glands also help in excretion. ✓

(1) Only (i)

(2) Both (ii) and (iii)

✓ (3) Both (i) and (iii)

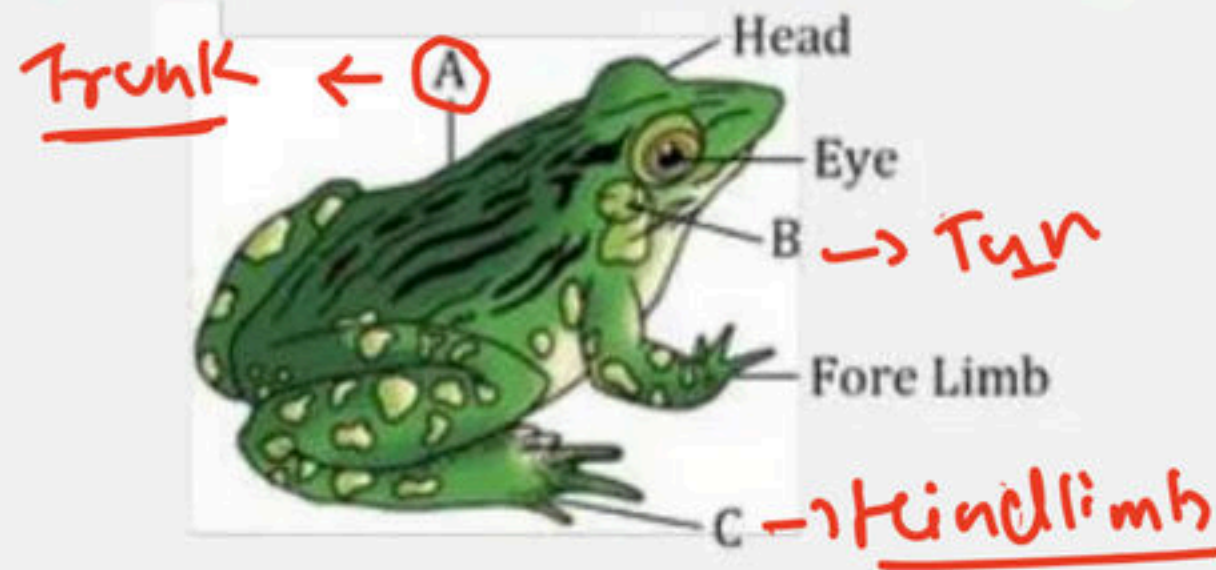
(4) All of these





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

88. See the following figure and identify A to C respectively



(1) Neck, Tympanum, Hind limb

☒ (2) Trunk, Tympanum, Hind limb

(3) Neck, Brown eye spot, Web

(4) Trunk, Tympanum, Web



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

89. Match the description given in column I with their examples given in column II and choose the correct option.

| Column I<br>(Description) |                                                                                               | Column II<br>(Example) |               |
|---------------------------|-----------------------------------------------------------------------------------------------|------------------------|---------------|
| (A)                       | Aquatic respiratory organ →                                                                   | (I)                    | Skin          |
| (B)                       | Organ which acts urogenital duct and opens into the cloaca →                                  | (II)                   | Ureter        |
| (C)                       | A small median chamber that is used to pass faecal matter, urine and sperms to the exterior → | (III)                  | Cloaca        |
| (D)                       | A triangular structure which joins the right atrium and receives blood through vena cava →    | (IV)                   | Sinus venosus |

✓ (1) A-I; B-II; C-III; D-IV

(2) A-II; B-IV; C-I; D-III

(3) A-II; B-I; C-III; D-IV

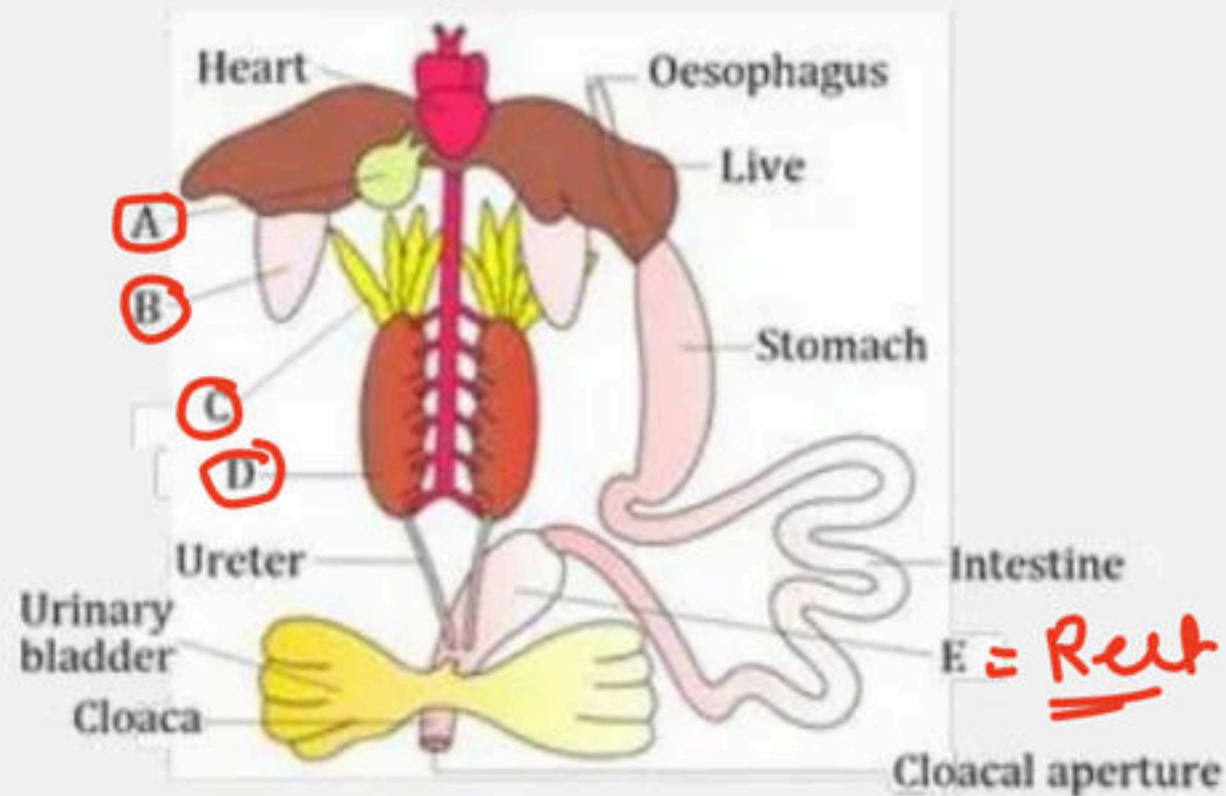
(4) A-IV; B-II; C-I; D-III





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

90. The given figure is related to diagrammatic representation of internal organs of frog. Identify A to E



|       | A            | B    | C          | D      | E      |
|-------|--------------|------|------------|--------|--------|
| ✓ (a) | Gall bladder | Lung | Fat bodies | Kidney | Rectum |
| (b)   | Gall bladder | Lung | Testis     | Kidney | Rectum |
| (c)   | Gall bladder | Lung | Fat bodies | Testis | Rectum |
| (d)   | Gall bladder | Lung | Ovary      | Testis | Rectum |





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

91. Assertion : Columnar epithelium lining the intestinal mucosa appears to have a brush like appearance.

Reason : A large number of microvilli are present on brush bordered columnar epithelium.

- ↓
- (1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion
  - (2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion
  - (3) If the assertion is true but the reason is false
  - (4) If both the assertion and reason are false





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

92. Which of the following statement(s) is/are correct about nervous system of cockroach? PYQs

✓ (i) It consists of a series of fused segmentally arranged ganglia joined by paired longitudinal connectives on the ventral side. ✓

(ii) There are six ganglia <sup>(3)</sup> lie in the thorax, and three in the abdomen. <sup>6</sup> ✗

✓ (iii) The sense organs are antennae, eyes, maxillary pulps, labial pulps and anal cerci etc.

(iv) Each eye consists of about ~~5000~~ 2000 hexagonal ommatidia.

✓ (1) Both (i) and (iii)

(2) Only (ii)

(3) Both (i) and (iv)

(4) All of these



## Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

93. The sinus venosus is located on

- (1) Dorsal surface of the heart of frog ✓
- (2) Ventral surface of the heart of frog
- (3) Dorsal surface of the heart of rabbit
- (4) Ventral surface of the heart of rabbit





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

94. Match the following simple epithelial tissues in column I with their occurrence in column II and choose the correct combination from the options given

| Column I             | Column II            |
|----------------------|----------------------|
| A. Squamous          | 1. Intestinal glands |
| B. Cuboidal          | 2. Trachea           |
| C. Columnar          | 3. Ovary             |
| D. Ciliated          | 4. Blood vessels     |
| E. Pseudo stratified | 5. Bronchioles       |

(1) A - 1, B - 2, C - 4, D - 3, E - 5

(2) A - 5, B - 4, C - 2, D - 1, E - 3

(3) A - 4, B - 5, C - 1, D - 2, E - 3

(4) A - 4, B - 3, C - 1, D - 2, E - 5



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

95. Find the wrongly matched pair

(1) Unicellular glandular cells – Goblet cell

(2) Saliva – Exocrine secretion

(3) Fusiform fibres – Smooth muscle

(4) Cartilage – Areolar tissue ✓

↓  
Special





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

96. Match column-I (type of epithelium) with column-II (Description) and choose the correct option.

| Column I<br>(Types of epithelium) |                     | Column II<br>(Description) |                                                                                                 |
|-----------------------------------|---------------------|----------------------------|-------------------------------------------------------------------------------------------------|
| (A)                               | Squamous epithelium | (I)                        | It is composed of a single-layer of cube-like cells                                             |
| (B)                               | Cuboidal epithelium | (II)                       | Having cilia on their free surface                                                              |
| (C)                               | Columnar epithelium | (III)                      | It is composed of a single layer of tall and slender cells                                      |
| (D)                               | Ciliated epithelium | (IV)                       | It is made up of a <u>single thin layer of flattened cells</u> with <u>irregular boundaries</u> |

(1) A-IV; B-I; C-III; D-II

(2) A-I; B-IV; C-III; D-II

(3) A-IV; B-I; C-II; D-III

(4) A-IV; B-III; C-I; D-II



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

97 In cockroach, excretion is brought about by: [NEET]

A. Phallic gland B. Urecose gland

C. Nephrocytes D. Fat body E. Collateral glands <sup>x</sup> <sub>→ fen</sub>

Choose the correct answer from the options given below :

(1) B and D only

(2) A and E only

(3) A, B and E only

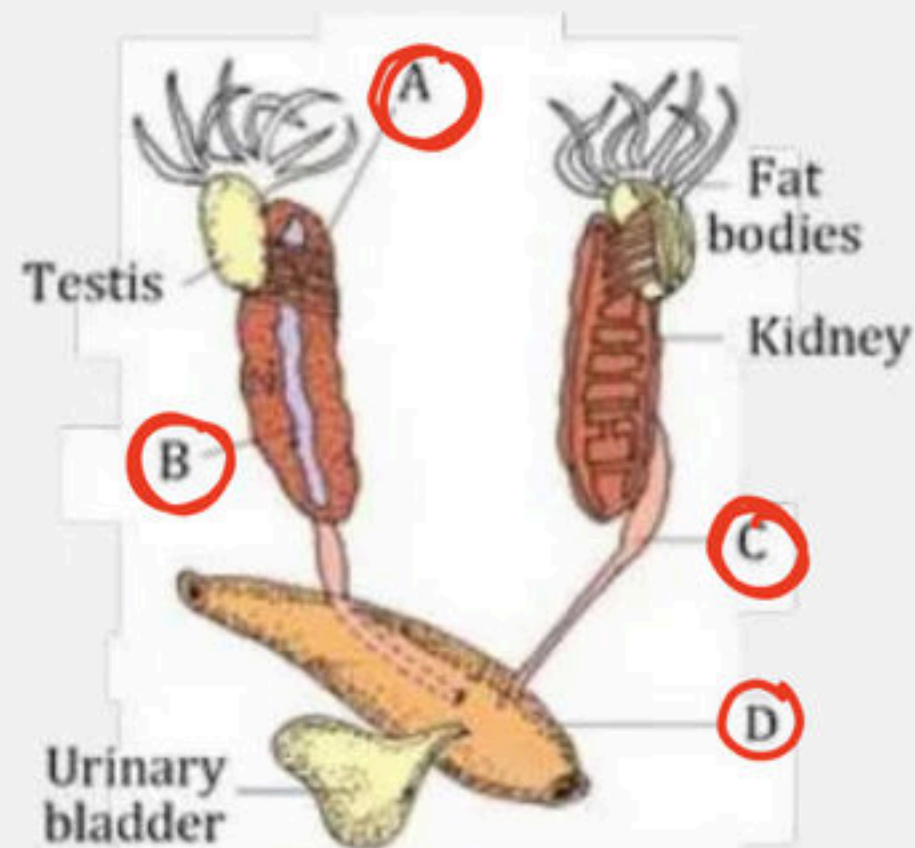
(4) B, C and D only





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

98. Observe the following figure indicating the male reproductive system of frog. Identify A, B, C and D



|      | A               | B             | C                 | D      |
|------|-----------------|---------------|-------------------|--------|
| (a)  | Vasa efferentia | Thyroid gland | Urinogenital duct | Cloaca |
| ✓(b) | Vasa efferentia | Adrenal gland | Urinogenital duct | Cloaca |
| (c)  | Bidder's canal  | Adrenal gland | Urinogenital duct | Cloaca |
| (d)  | Bidder's canal  | Adrenal gland | Urinogenital duct | Rectum |



# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

99. The cloaca in frog is a common chamber for the urinary tract, reproductive tract and:



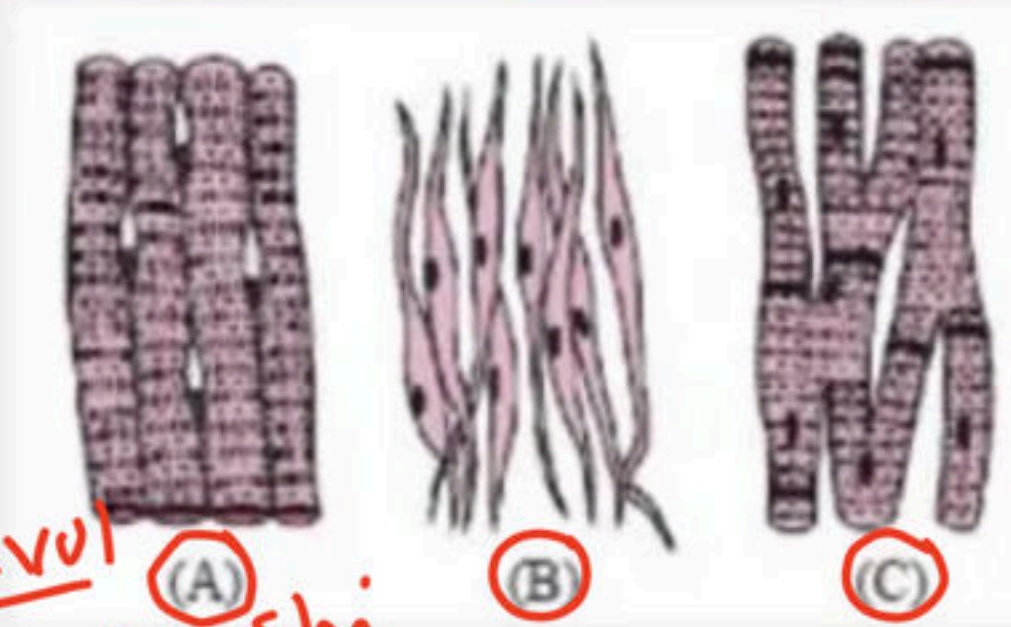
- ✓✓ (1) Alimentary canal
- (2) Portal system
- (3) Hepatic portal vessels
- (4) Notochord





# Title: Structural Organisation in Animal (Tissue+Frog+Cockroach)

100. The following figures A, B and C are types of muscle tissue. Identify A, B and C.



(1) A-Smooth muscle, B-Cardiac muscle, C-Skeletal muscle

(2) A-Skeletal muscle, B-Smooth muscle, C-Cardiac muscle

(3) A-Cardiac muscle, B-Smooth muscle, C-Skeletal muscle

(4) A-Smooth muscle, B-Skeletal muscle, C-Cardiac muscle

voluntary  
striated

involuntary  
striated

involuntary  
striated





THANK YOU



**BIOLOGYLIVE**

cell: The unit of life

- Sheet

- cell: the unit of life

Thank you =>







▲ 2 · Asked by Hitesh

Sir 1st day aapne kon si app k bareme kaha tha